
Study of adhesin-substrate interactions at the molecular level for the development of better anti-bacterial materials

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Abstract - Post-surgical infections due to biofilm formation on implants and implant-based medical devices are a major cause of implant failure and increased healthcare costs. *Staphylococcus aureus* is responsible for up to 58% of periprosthetic total knee arthroplasty (TKA) infections, and accounts for 55% of orthopedic implant infections in India. To establish a robust computational framework for investigating pathogen-biomaterial interactions, this study first focuses on molecular dynamics (MD) simulations of ubiquitin, a well-characterized model protein, in various aqueous environments, including water sphere, water box, and generalized Born implicit solvent. System assembly, energy minimization, equilibration, and trajectory analysis were performed using NAMD as the simulation engine and VMD for visualization and analysis. The reliability and accuracy of the simulation setup were validated by monitoring structural and energetic parameters such as RMSD, energy distribution, and force profiles, which closely matched reference data. The hydroxyapatite (HA) supercell surface was modeled using pymatgen and ASE to generate realistic, simulation-ready substrates for future studies. This foundational work establishes validated protocols and system preparation strategies, providing a reliable platform for subsequent investigations of *S. aureus* adhesin interactions with HA surfaces, with the ultimate goal of guiding the design of anti-biofilm implant materials.

Keywords: *adhesion forces, hydroxyapatite, molecular dynamics, Staphylococcus aureus, ubiquitin*

1 INTRODUCTION

A study at Mandya Institute of Medical Sciences (Karnataka) reported a surgical site infection (SSI) rate of 4.43% within 3 months of orthopedic implant surgery. Most infections were deep (81.8%), and *Staphylococcus aureus* was the most common pathogen^[1]. A prospective study in South India found that methicillin-resistant *Staphylococcus aureus* (MRSA) was the most common pathogen in orthopedic surgical site infections, accounting for 48.4% of isolates^[2]. Hydroxyapatite being chemically similar to the mineral component of natural bone, makes it highly compatible with human tissue and minimizes the risk of rejection or adverse reactions^[3]. Hence, hydroxyapatite is widely used as a coating for bone implants^[4]. *S. aureus* can readily attach to implant surfaces and form biofilms, which are structured communities of bacteria encased in a self-produced extracellular matrix^[5]. The matrix limits the penetration of antimicrobial agents and shields bacteria from immune cells, making infections very difficult to eradicate^[6]. Adhesion is the first step in biofilm formation, which makes infections persistent and difficult to treat^[7]. Simulation and quantification of how *S. aureus* adheres to HA can guide the design of implant surfaces that resist bacterial colonization, supporting bone integration.

However, the main challenges to this objective stem from the biological complexity of *S. aureus*, making it extremely tough to accurately model all the components that possibly contribute to adhesion, and the lack of comprehensive reference data, which make NAMD simulations of this organism computationally demanding^[8]. Prior to modelling *S. aureus*, a preliminary study needs to be conducted on a simpler system to establish a foundational understanding of the host cell's protein regulation machinery^[9]. Ubiquitin is present in virtually all eukaryotic cells and is involved in a wide array of cellular processes, making it an ideal model system for studying protein regulation^[10].

The primary goal was to look upon conformational dynamics and stability in a biologically relevant environment^[11]. The protein needs to be solvated, i.e., put inside water, to more closely resemble the cellular environment. Hence, ubiquitin was put in a water sphere, to simulate protein in a minimally solvated, non-periodic environment, a water box, to mimic the bulk aqueous environment found in cells that allows for realistic long-range electrostatic interactions using methods like Particle Mesh Ewald (PME) and a generalized Born implicit solvent (GBIS), that models the effect of water without explicitly simulating water molecules, significantly reducing computational cost^[12].

NAMD was used as the engine for Molecular Dynamics (MD) because of its capability to handle large bio-molecular systems. NAMD allows multi-core processing which makes simulations run faster than they can on MD engines such as AMBER or GROMACS^[13]. NAMD also offers support for both explicit (water sphere, water box) and implicit (generalized Born) solvent models and implements advanced algorithms for evaluating electrostatic interactions (such as Particle Mesh Ewald) and uses multiple-time-stepping integration which increases computational efficiency. Visualization of all the simulation trajectories were performed using VMD. Structure preparation, solvation, equilibration, energy conservation, and mechanical response were first performed on ubiquitin and close agreement of statistical data- RMSD, energy distribution, force curve- made it evident that the system used was accurate and reliable. This provided a strong foundation for subsequent simulations of *Staphylococcus aureus* on Hydroxyapatite.

Building on these validated protocols, the next phase involved constructing atomistic models of the hydroxyapatite (HA) surface using force fields that accurately represent chemical bonding. The hydroxyapatite (HA) supercell surface was modeled using pymatgen, a robust Python library for materials analysis and structure manipulation^[14]. Starting from the bulk crystal structure of hydroxyapatite, pymatgen's transformation tools were used to generate a surface slab with a specified Miller index, ensuring the correct crystallographic orientation for subsequent simulations. The ASE (Atomic Simulation Environment) library was then utilized to further manipulate the atomic structure, including adding vacuum layers and adjusting slab dimensions.

These initial simulations not only validated the reliability of the computational setup but also established essential protocols for system preparation, analysis, and interpretation.

2 MATERIALS AND METHODS

All molecular dynamics were carried out using NAMD 3.0.1(Linux-x86_64-multicore-CUDA), which enables efficient large-scale simulations using GPU acceleration^[15]. NVIDIA GPUs using CUDA version 12.8 were used to leverage parallel processing for better computation. The CHARMM27 force field was used by the NAMD (Nanoscale Molecular Dynamics) to model all interactions. It includes CMAP correction, which improves protein backbone behavior and thus yields more accurate dynamic properties of the protein. VMD(Visual Molecular Dynamics) version 1.9.3, was used for trajectory visualization, molecular rendering, and basic post-simulation analysis^[16].

2.1 System Setup of Ubiquitin

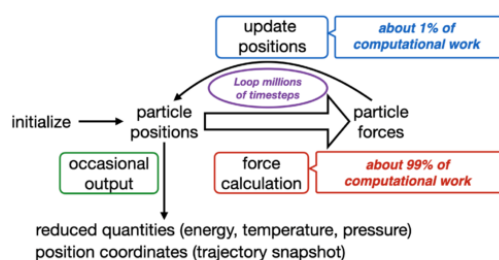


Figure 1: Workflow of running a molecular dynamics (MD) simulation using NAMD

The initial ubiquitin pdb file named 1UBQ.pdb was obtained from the protein data bank. The CHARMM27 parameter and topology files used in this study were `par_all27_prot_lipid.inp` and `top_all27_prot_lipid.inp`. These provide CHARMM27 force field data tailored for proteins and lipids. To obtain the psf file of ubiquitin, the psfgen package of VMD was used. A pgn file named `ubq.pgn` was created, which included commands to target the psfgen and guess coordinates of missing atoms (like hydrogens) based on residue definitions from the topology files.

2.2 Solvating Ubiquitin

The psf file obtained above has hydrogen atoms but no water molecules. The protein needs to be solvated in two different environments (a water sphere and a water box) to resemble the cellular environment and to study boundary effects.

The water sphere environment was prepared for simulating minimization and equilibration of the protein without periodic boundary conditions. The `wat_sphere.tcl` script in VMD was used for embedding the protein in a spherical droplet of TIP3P water^[18]. The radius and center of the water sphere were automatically calculated for applying spherical boundary conditions and electrostatic treatments using the Multilevel Summation Method (MSM) in NAMD (refer to Figure 2(b)).

For periodic boundary condition simulations, the protein was solvated in a cubic water box using the solvate plugin in VMD with a 5 Å buffer from the protein surface in all directions. This configuration takes isotropic pressure coupling and Particle Mesh Ewald (PME) for long-range electrostatics into account. The center of the water box was measured using VMD's `measure center` command and it was used to define the periodic cell origin in the simulation input.

2.3 Minimization and Equilibration of the Ubiquitin-Water Setup

Minimization of the protein structure was done in order to ensure the protein was in a local minimum on the potential energy surface, free from steric or atomic clashes which hence gave a physically stable basis to the net forces on the protein before proceeding with the simulations^[19]. The water box and the water sphere systems were subjected to 100 steps of energy minimization and all atomic velocities were set to zero prior to the equilibration step. The system was then gradually heated to the desired temperature (310 K) and brought to thermodynamic equilibrium at the same, using a controlled Langevin thermostat^[20]. Each equilibration simulation goes for 2500 steps (5 ps), and trajectory data were saved for every 100 steps. The same was done with a Generalized Born Implicit Solvent (GBIS). An implicit solvent model is a simulation technique which eliminates the need for explicit water atoms by including many of the effects of solvent in the inter-atomic force calculation. The configuration file taking the above details into account was run on NAMD.

2.4 RMSD Analysis of Ubiquitin Water Sphere

RMSD (Root Mean Square Deviation) analyzes the extent to which the protein system has equilibrated. RMSD characterizes how much the system deviated from its initially defined position in space. The simulation trajectory file generated after the minimization and equilibration run (`ubq_ws_eq.dcd`) was loaded along with the structure files (`ubq_ws.psf`) and initial coordinates (`ubq_ws.pdb`). Only backbone atoms (N, C α , C) were selected to eliminate contributions from side-chain fluctuations, for RMSD calculations^[20].

2.5 Maxwell-Boltzmann Energy Distribution

Kinetic Energy distribution plot is constructed for the equilibrated ubiquitin in water sphere system. The velocities of the atoms were extracted from the trajectory file (`ubq_ws_eq.dcd`) using VMD. Per-atom kinetic energies were calculated using the classical relation:

$$E_k = \frac{1}{2}mv^2$$

where m is the atomic mass and v is the velocity vector of the atom. The kinetic energies of the equilibrated atoms were binned to generate a histogram and compared against the theoretical Maxwell-Boltzmann distribution for a system at 310 K:

$$P(E_k) = \frac{2}{\sqrt{\pi}} \left(\frac{1}{k_B T} \right)^{3/2} \sqrt{E_k} \exp \left(-\frac{E_k}{k_B T} \right)$$

where k_B is Boltzmann's constant and T is the temperature. The empirical distribution was normalized and plotted to verify consistency with the expected thermal behavior under Langevin dynamics. Xmgrace was used for the visualization segment.

2.6 Temperature Autocorrelation Function

To assess the ubiquitin system's relaxation dynamics, the time correlation of temperature fluctuations was analyzed. The temperature autocorrelation function $C(\tau)$ was constructed by computing the instantaneous temperature from the total kinetic energy at each frame of the trajectory. It is defined as:

$$C(\tau) = \langle (T(t) - \langle T \rangle)(T(t + \tau) - \langle T \rangle) \rangle$$

where $\langle T \rangle$ is the mean temperature over the trajectory. A custom Tcl script in VMD was used to compute the autocorrelation, which was then normalized such that $C(0) = 1$. The decay profile of $C(\tau)$ characterizes the timescale over which the system loses memory of its thermal fluctuations and approaches equilibrium. This also reflects the effectiveness of the Langevin thermostat in damping thermal noise. The results were visualized using XMGrace for temporal analysis.

2.7 Steered Molecular Dynamics (SMD)

Steered Molecular Dynamics (SMD) is a powerful computational technique used to investigate the elastic properties, unfolding mechanisms, and conformational transitions of biomolecules at atomic resolution^[22]. In SMD, an external force is systematically applied to a selected group of atoms—referred to as SMD atoms—while another set is held fixed. This configuration allows the simulation of mechanical processes such as protein unfolding, ligand unbinding, and conformational changes, closely mimicking experimental techniques like atomic force microscopy (AFM) (refer to Figure 8).

2.7.1 Constant Velocity Pulling

SMD simulations driven by constant velocity pulling were conducted with NAMD and the CHARMM27 force field to explore the mechanical unfolding of ubiquitin. The system was solubilized in a TIP3P water box and represented using the files `ubq_wb.psf` and `ubq_wb.pdb`. A harmonic spring with a stiffness of $7.0 \text{ kcal/mol/\AA}^2$ was connected to the C-terminal atom, which was drawn along the z-axis at a speed of $0.1 \text{ \AA/ps}$, while the N-terminal atom remained fixed with harmonic constraints. Simulations were performed in the NPT ensemble at 310 K and 1 atm, utilizing Langevin dynamics for temperature control (damping coefficient of 1 ps^{-1}) and the Langevin piston method to maintain pressure. Long-range electrostatics were addressed using the Particle Mesh Ewald (PME) approach, with a grid spacing of about $1.0 \text{ \AA}$. Nonbonded interactions used a cutoff distance of $12 \text{ \AA}$ and a switching function beginning at $10 \text{ \AA}$ to gradually reduce van der Waals forces. The SHAKE algorithm was used to constraint all bonds containing hydrogen atoms, allowing for a 2 fs integration time step. Periodic boundary conditions were implemented in all three dimensions. Regular intervals produced simulation output, comprising trajectory (.dcd), energy (.log), and restart files, which were subsequently post-processed to derive force–extension profiles and evaluate unfolding dynamics.

2.7.2 Constant Force Pulling

Simulations with constant force pulling were conducted using NAMD and the CHARMM27 force field to study how ubiquitin reacts to prolonged mechanical stress. The system, immersed in a TIP3P water box and characterized by `ubq_wb.psf` and `ubq_wb.pdb`, was exposed to a steady force of $5.0 \text{ kcal/mol/\AA}$ directed at the C-terminal atom along the z-axis, with the N-terminal atom held in place. Instead of using a harmonic spring for constant velocity pulling, the force was applied directly as outlined in `ubq_cforce.conf`. Simulations were conducted in the NPT ensemble at 310 K and 1 atm, employing Langevin dynamics (damping coefficient 1 ps^{-1}) along with the Langevin piston technique. Electrostatics were handled with PME (grid spacing $\sim 1.0 \text{ \AA}$), and a cutoff of $12 \text{ \AA}$ along with a switching distance of $10 \text{ \AA}$ was applied for nonbonded interactions. Hydrogen bonds were limited using SHAKE, and a 2 fs time step was employed. Periodic boundary conditions were implemented. The 2 ns simulation produced trajectory (.dcd) and log (.log) files, which were used to analyze force–extension behavior and mechanical work.

2.8 Modelling the Hydroxyapatite surface

To model the hydroxyapatite (HA) substrate, we constructed a 001 surface slab using the crystal structure data from a CIF file. The slab was generated by cleaving the bulk HA along the (0,0,1) Miller plane to expose the biologically relevant 001 facet. A two-layer thick slab was created and replicated 8×7 times along the X and Y directions, respectively, to produce a sufficiently large surface area for subsequent protein–substrate interaction studies. A 20 Å vacuum was introduced along the Z-axis to prevent artifacts from periodic boundary conditions and to mimic a vacuum environment, as used in simulations such as those reported in Raj et al. (2020)^[21].

2.9 Modelling SpA and Hydroxyapatite environment

The B-domain of *Staphylococcus aureus* protein A (SpA), recognized for its ability to adhere to tissues and implants, was utilized to investigate protein–material interactions. The atomic structure of SpA was sourced from the Protein Data Bank (PDB ID: 2JWD) and saved as `spa_ready.pdb` following preprocessing that involved eliminating unnecessary chains and aligning it along the Z-axis so that the binding domain was directed towards the hydroxyapatite surface.

The hydroxyapatite (HA) substrate was created using the crystallographic information from the file `ha_mp4019.cif`, obtained from the Materials Project database. The `pymatgen` library was utilized to transform this CIF file into a periodic supercell showcasing the {001} surface, which was subsequently saved as `ha_supercell.pdb`. The HA supercell was built with lateral dimensions large enough to fit the entire lateral span of the SpA domain while avoiding interactions with periodic images. To effectively characterize the hybrid system comprising biomolecule and ceramic substrate, we utilized a mix of the CHARMM27 force field (for the SpA protein) and the INTERFACE force field (for HA). The INTERFACE force field, created by Heinz and colleagues, allows dependable atomistic modeling of inorganic solids and surfaces, and has been modified into a CHARMM-compatible format. An interface file tailored for custom use, consisting of a residue topology file (`interface_ha.rtf`) and a parameter file (`interface_ha.prm`), was required to model hydroxyapatite. These files specified atom types, partial charges, bonding parameters, and Lennard-Jones van der Waals parameters for calcium (Ca^{2+}), phosphate (PO_4^{3-}), and hydroxyl (OH^-) groups.

The SpA–HA system was formed by positioning `spa_ready.pdb` above `ha_supercell.pdb` with an approximate vertical distance of ~ 5 Å initially. The combined structure was stored as `merged_ready.pdb`. To study intrinsic unfolding behavior free from substrate effects, the SpA protein was immersed in a spherical TIP3P water shell via VMD’s `solvate` plugin. Solvating SpA within a water sphere isolates the protein’s inherent mechanical response without influence from substrates, simulating physiological hydration.

3 Results and Discussion

3.1 System Setup

`ubq.pdb` and `ubq.psf` were obtained after `ubq.pgn` was run on VMD. `ubq.pdb` (Figure 2(a)) file contains the complete coordinates of all atoms, including hydrogens. `ubq.psf` has the complete structural information of the protein.

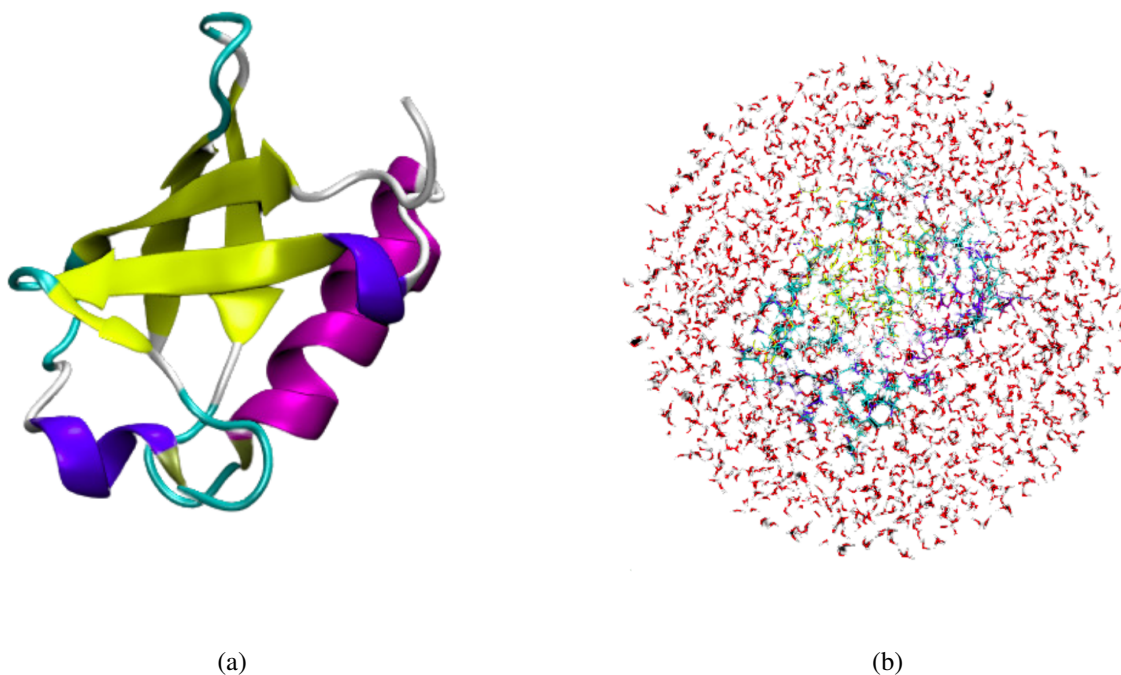


Figure 2: Ubiquitin structures, as visualised on VMD (a) Ubiquitin molecule in 3D ribbon format. (b) Ubiquitin molecule in a water sphere.

3.2 Root Mean Square Deviation (RMSD)

RMSD of the ubiquitin system rised to $0.842 \text{ \AA}$ and then plateaus after a value $\sim 0.71 \text{ \AA}$, showing structural deviation over time. Low RMSD indicates protein majorly upholds its overall structure without scope for degradation^[23]. Xmgrace was used to analyze the `.dat` file, which signified the time series and other numerical analysis data, made after running the configuration file. The initial peak in the plot indicates the increase in energy as it moves away from its minimum potential state. The fluctuations indicate the minor conformational changes gone through by the entire protein. Ultimate flattening out of the curve indicates the fluctuation around an equilibrium state (refer to Figure 3).

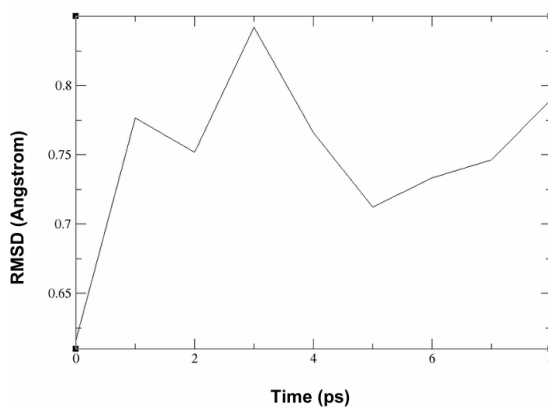


Figure 3: RMSD (Angstrom) vs Time (ps) for the entire protein.

3.3 Maxwell Boltzmann distribution of KE

The Kinetic energy distribution spans from 0 to 10 kcal/mol, but over 90% of the ubiquitin atoms have Kinetic energies below 2 kcal/mol, showing a steep decay in frequency as energy increases. Another output of the equilibration and minimization run was `ubq_ws_eq_1fs.restart.vel` file which specified the velocities of all atoms at a particular simulation step. The kinetic energies of all these atoms taking into account these velocities were calculated and run and compared to the Maxwell-Boltzmann kinetic energy distribution^[24]. The Maxwell-Boltzmann distribution is the expected statistical behavior for a system at thermal equilibrium at a given temperature. It states the expected probability distribution for kinetic energies of a particle at a fixed temperature is such that most of them have lower kinetic energies and fewer have high kinetic energies. The shape of the graph indicated the system behaved as expected at thermal equilibrium (refer to Figure 4).

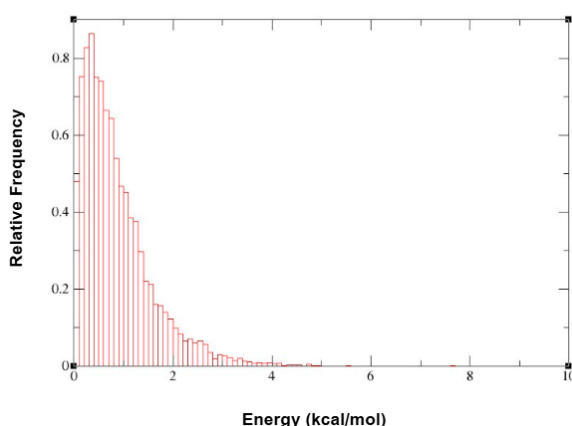


Figure 4: Maxwell-Boltzmann Distribution of KE among particles (Relative Frequency vs Energy (kcal/mol)).

3.4 Temperature Autocorrelation Function (TACF)

The autocorrelation function drops from 1.0 to ~ 0.2 within the first 5 fs, indicating a rapid loss of temperature correlation. The Temperature Autocorrelation function gives the measure of how the temperature at a certain time is correlated with its value at a later time. The red curve is a decaying exponential fit. Initially, temperature is strongly correlated with itself. However, this drops off quickly, since the actual data points (black curve) show thermal fluctuations and ultimately the correlation function oscillate around 0, suggesting the thermal fluctuations dominate (refer to Figure 5).

3.5 SMD : Constant Force Pulling

The N-C terminal separation rises from 40.7 Å to 45.1 Å over 35 time steps, indicating a net elongation of 4.4 Å. The trajectory of ubiquitin under constant force pulling reveals a stepwise increase in the N–C terminal separation, as shown in the first graph. The presence of multiple plateaus in the separation profile suggests that the protein unfolds through a series of stable or metastable intermediate states. Each plateau represents a structural configuration in which the applied force is insufficient to induce further unfolding until a critical threshold is overcome. This behavior indicates that the protein resists deformation until specific intramolecular interactions—such as hydrogen bonds or hydrophobic packing—are disrupted. Such discrete unfolding events are characteristic of globular proteins like ubiquitin, which possess well-defined secondary and tertiary structures (refer to Figure 6).

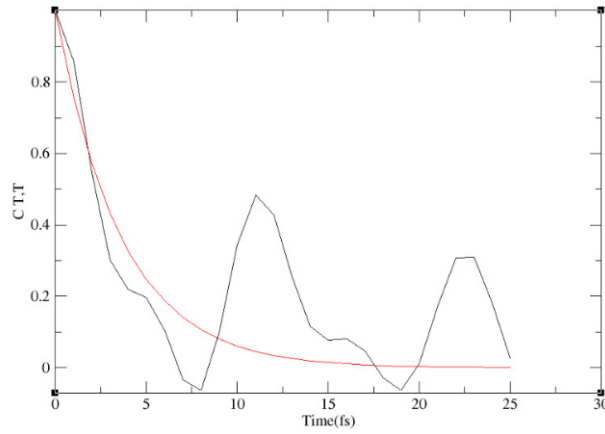
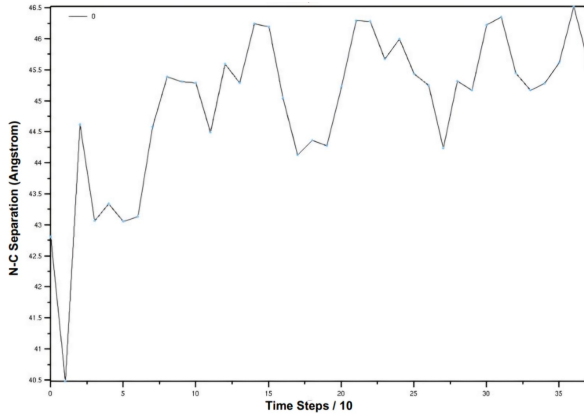


Figure 5: Correlation Function vs Time (fs) fit to a decaying approximation.



(a)

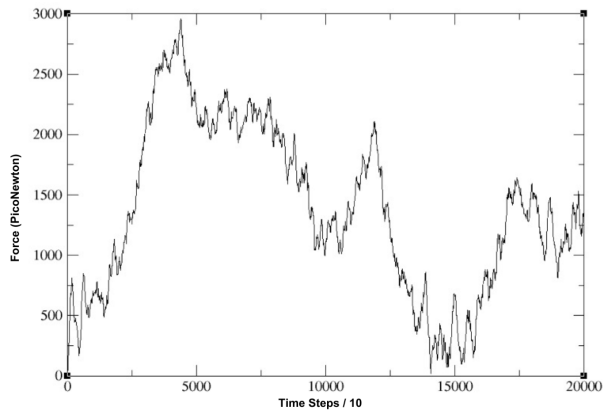
Timestep / 10	N-C Distance (Å)
0.0	40.2
1.0	44.5
2.0	42.7
3.0	43.0
4.0	43.2
5.0	45.0
6.0	44.8
7.0	45.3
8.0	43.8
9.0	44.9
10.0	45.6
11.0	44.7
12.0	45.0
13.0	44.3
14.0	44.9
15.0	45.9
16.0	44.0
17.0	44.2
18.0	44.7
19.0	45.5
20.0	44.6
21.0	45.1
22.0	45.8
23.0	43.9
24.0	44.1
25.0	43.5
26.0	44.7
27.0	45.6
28.0	45.4
29.0	44.3
30.0	45.7
31.0	44.1
32.0	45.9
33.0	44.0
34.0	46.0
35.0	44.5

(b)

Figure 6: Response of ubiquitin under constant force pulling simulation. (a) Time evolution of N-C terminal separation in ubiquitin. (b) Tabulated N-C terminal distances over time.

3.6 SMD : Constant velocity pulling

The applied force reaches a high of about 2636 pN near timestep 4000, followed by several local maxima and minima, signifying a sequence of unfolding occurrences. The force decreases to 318.03 pN at timestep 14000, indicating structural relaxation following the initial unfolding. In the constant velocity SMD simulation, the force experienced by the protein exhibits a saw-tooth profile, with a sequence of sharp peaks followed by rapid drops in force. Each peak corresponds to a significant unfolding event, such as the rupture of a β -strand or a domain, requiring a substantial force to overcome. The subsequent force drop reflects a relaxation period as the protein extends and offers less resistance. Over time, the overall trend shows a decrease in peak magnitudes, indicating progressive unfolding and loss of structural integrity. This pattern aligns with previously reported force-extension curves for ubiquitin and supports the view that mechanical unfolding involves sequential domain-level disruptions, each demanding variable energy depending on the structural stability of the segment involved^[14] (refer to Figure 7).



(a)

Timestep / 10	Force (pN)
0.0	0.0
1000.0	665.59
2000.0	985.39
3000.0	2007.26
4000.0	2636.27
5000.0	2173.7
6000.0	2256.32
7000.0	2261.11
8000.0	2019.59
9000.0	1588.69
10000.0	1114.73
11000.0	1321.42
12000.0	1863.02
13000.0	811.51
14000.0	318.03
15000.0	656.38
16000.0	546.67
17000.0	1422.08
18000.0	1523.53
19000.0	862.75
20000.0	1348.9

(b)

Figure 7: Force response of ubiquitin under constant velocity pulling simulation. (a) Force-extension plot from constant velocity pulling. (b) Extracted data values from force profile.

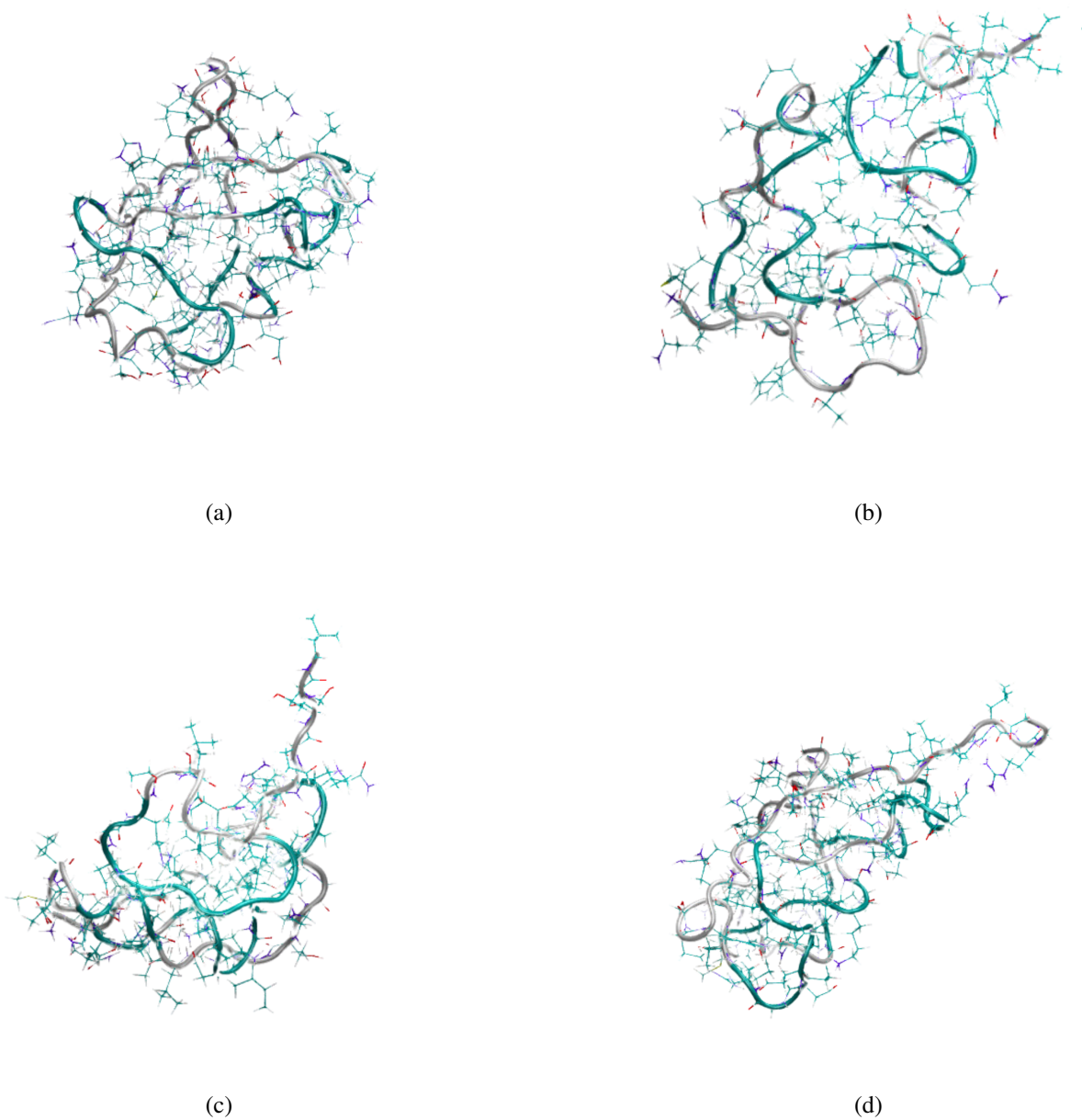


Figure 8: Structures (a)–(d) demonstrate the gradual unfolding of ubiquitin in SMD simulations conducted in NAMD. These frames were obtained via VMD from the `.dcd` trajectory file. The pictures illustrate different phases of structural changes as force is exerted at the protein's ends, emphasizing the progressive breakdown of secondary structure and molecular stretching.

3.7 Hydroxyapatite over VMD

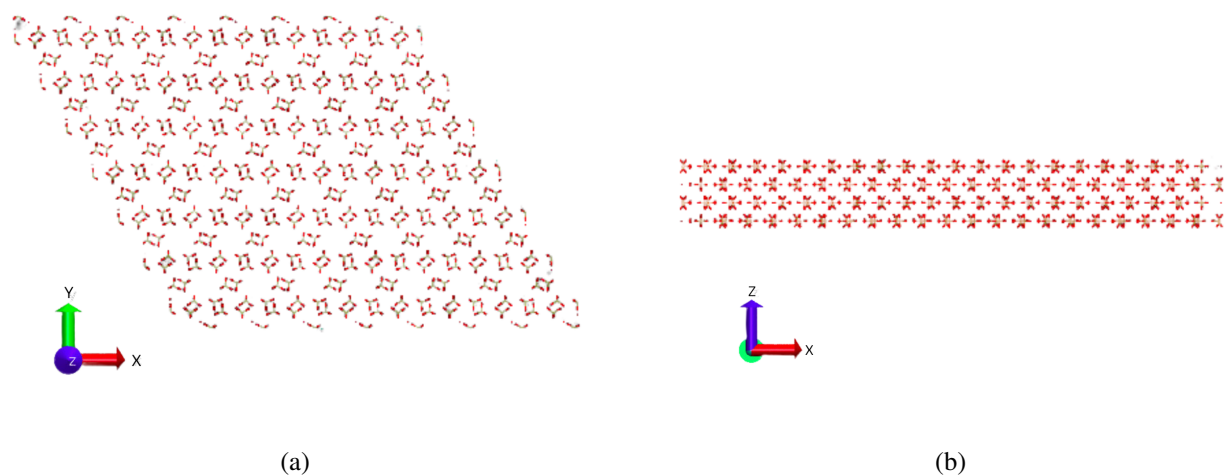


Figure 9: Visual representation of the hydroxyapatite supercell constructed for molecular dynamics simulations. (a) Side view showing lattice arrangement along the Y–Z plane. (b) Top view illustrating periodicity and atomic organization across the X–Y plane.

The hydroxyapatite (HA) 001 surface slab was constructed using the Pymatgen and the ASE (Atomic Simulation Environment) libraries. The initial .cif file for Hydroxyapatite was cleaved along the (001) plane^[25].

(Figure 9 (a)) showed the side view along the Y-Z plane thus confirming the presence of two atomic layers in the slab. The bilateral thickness and overall structural integrity ensure mechanical stability while avoiding surface distortions.

(Figure 9 (b)) showed the top-down view where the supercell had expanded 8×7 times and demonstrated periodicity and uniformity. This extended surface area is critical for minimizing boundary effects during subsequent simulations of microbial adhesion.

This well-defined and validated HA model could now be served as a reliable substrate for the next stage of our research, which involves modeling *Staphylococcus aureus* adhesion.

3.8 Modelling SpA and Hydroxyapatite environment

A major obstacle to run the *Staphylococcus aureus* and Hydroxyapatite simulation was the procurement of the interface files. SpA's PDB file and Hydroxyapatite's supercell CIF file were prepared, and they were merged into the same environment. But for preparing the PSF of the merged system, interface files of Hydroxyapatite were required for the psfgen. While CHARMM27 is widely used as a force field for biological systems, an interface file is required to effectively be able to model the interactions of inorganic substrates. However, the parameter files for constructing the interface files weren't available as an open source file and given the time constraint of 2 months, the same couldn't be arranged within the given time period. Hence, while Hydroxyapatite and SpA could be brought in the same environment, no simulation could be run for the system (Figure 10).

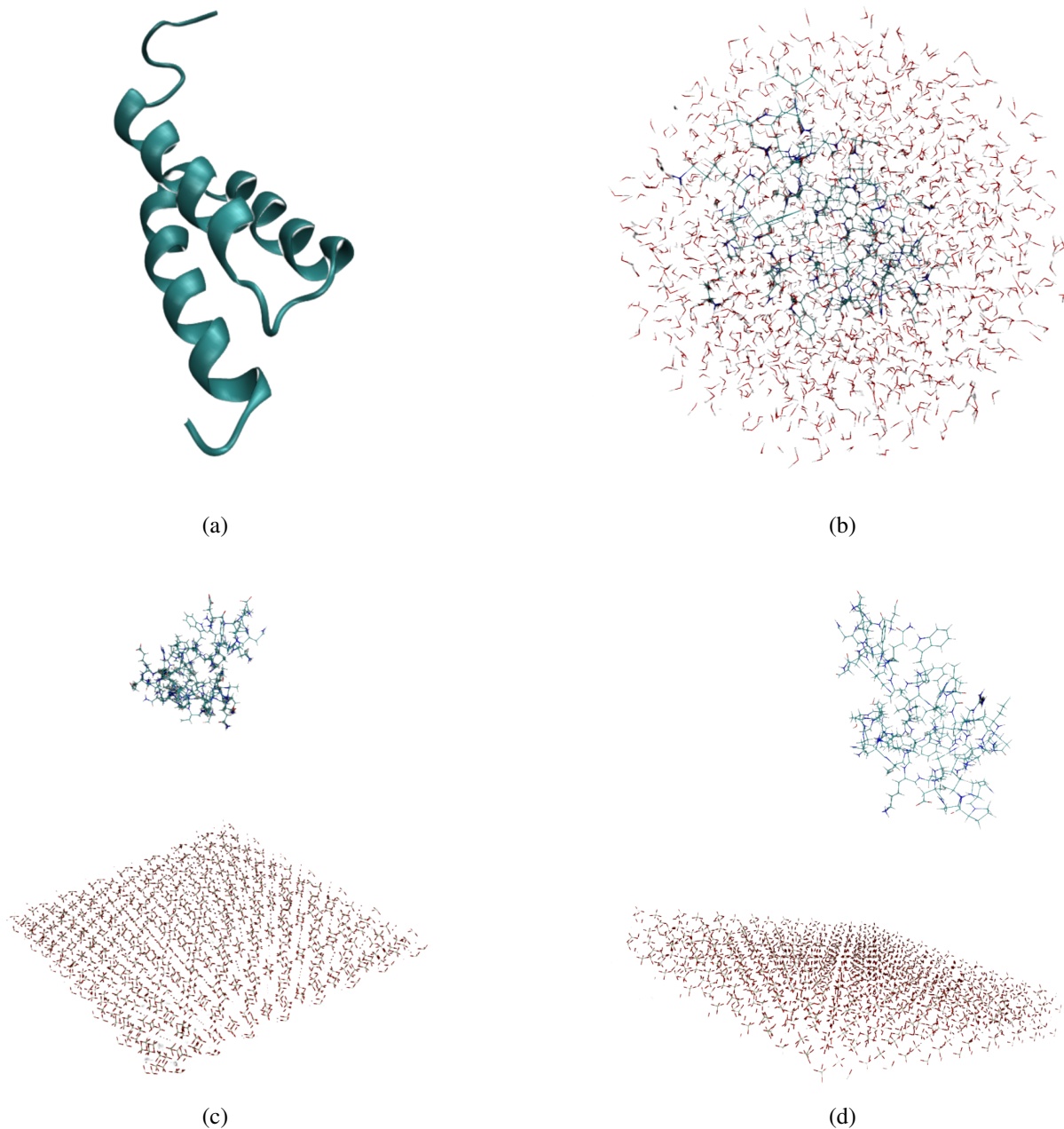


Figure 10: Structural representations of the *Staphylococcus aureus* surface protein A (SpA) in various simulation environments. (a) Isolated SpA protein in its native folded conformation. (b) SpA embedded in a solvated water sphere, illustrating the hydration shell used in preliminary equilibration. (c) SpA positioned 5 Å above the hydroxyapatite (HA) (001) surface slab, showing the full system used for protein–substrate interaction modeling. (d) Zoomed-in view of the SpA–HA interface at 5 Å distance, highlighting atomic-level proximity and orientation relevant for initial adhesion studies.

4 CONCLUSION

In this study, a molecular dynamics protocol was established using ubiquitin as a model protein for system preparation, equilibration simulations and analysis methodologies. Using NAMD for simulations and VMD for visualisation, trajectory analysis was conducted across explicit and implicit solvent models. The hydroxyapatite (HA) supercell surface was meticulously constructed using pymatgen and ASE, providing a realistic and customizable substrate for future interfacial studies. The results showed close agreement with reference structural and energetic parameters that confirmed reliability of computational framework.

Fundamental expertise in molecular dynamics configuration and evaluation facilitated precise modeling of adhesin attachment to hydroxyapatite, permitting thorough examination of interaction forces and structural reactions at the biointerface.

This confirmed the reliability of the approach for future protein-surface simulations and also paved the way for the ultimate objective: to be able to simulate the molecular mechanics of *Staphylococcus aureus* adhesin interactions with HA. Further insights gained from *S. aureus* and HA simulations will aid in understanding bacterial adhesion and guiding the rational design of anti-biofilm implant coatings.

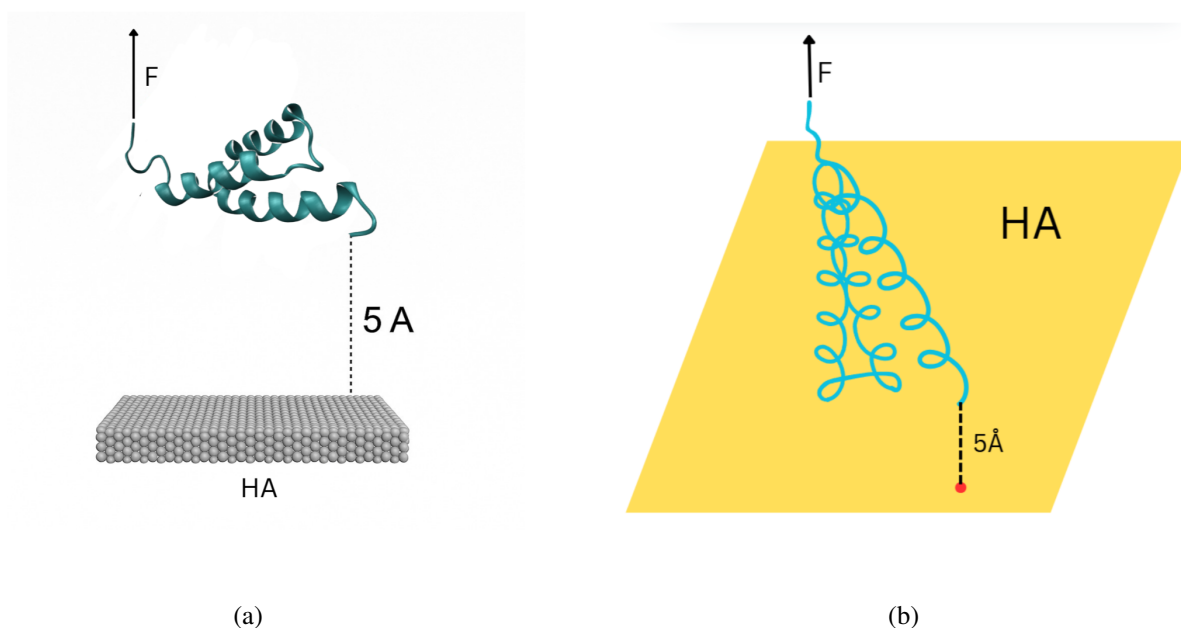


Figure 11: Conceptual schematics illustrating the SMD setup for investigating the adhesion behavior of *Staphylococcus aureus* protein A (SpA) on a hydroxyapatite (HA) surface. (a) Macro-level representation of the SMD process, depicting the application of a pulling force (F) on the SpA protein initially positioned $5 \text{ \AA}$ above the HA substrate. (b) Zoomed-in schematic detailing the initial configuration at the atomic scale, highlighting the orientation and proximity of SpA relative to the HA (001) surface prior to force application

Our goal is to investigate protein adhesion on biomaterial surfaces through steered molecular dynamics (SMD). The SpA is first positioned $5 \text{ \AA}$ above HA to prevent steric clashes while enabling nonbonded interactions to develop during minimization. One side of the protein is subsequently lifted while the substrate atoms remain secured, offering a stable reference. This arrangement allows accurate assessment of adhesion strength and unfolding characteristics, mimicking the detachment of proteins from material surfaces when subjected to mechanical stress.

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